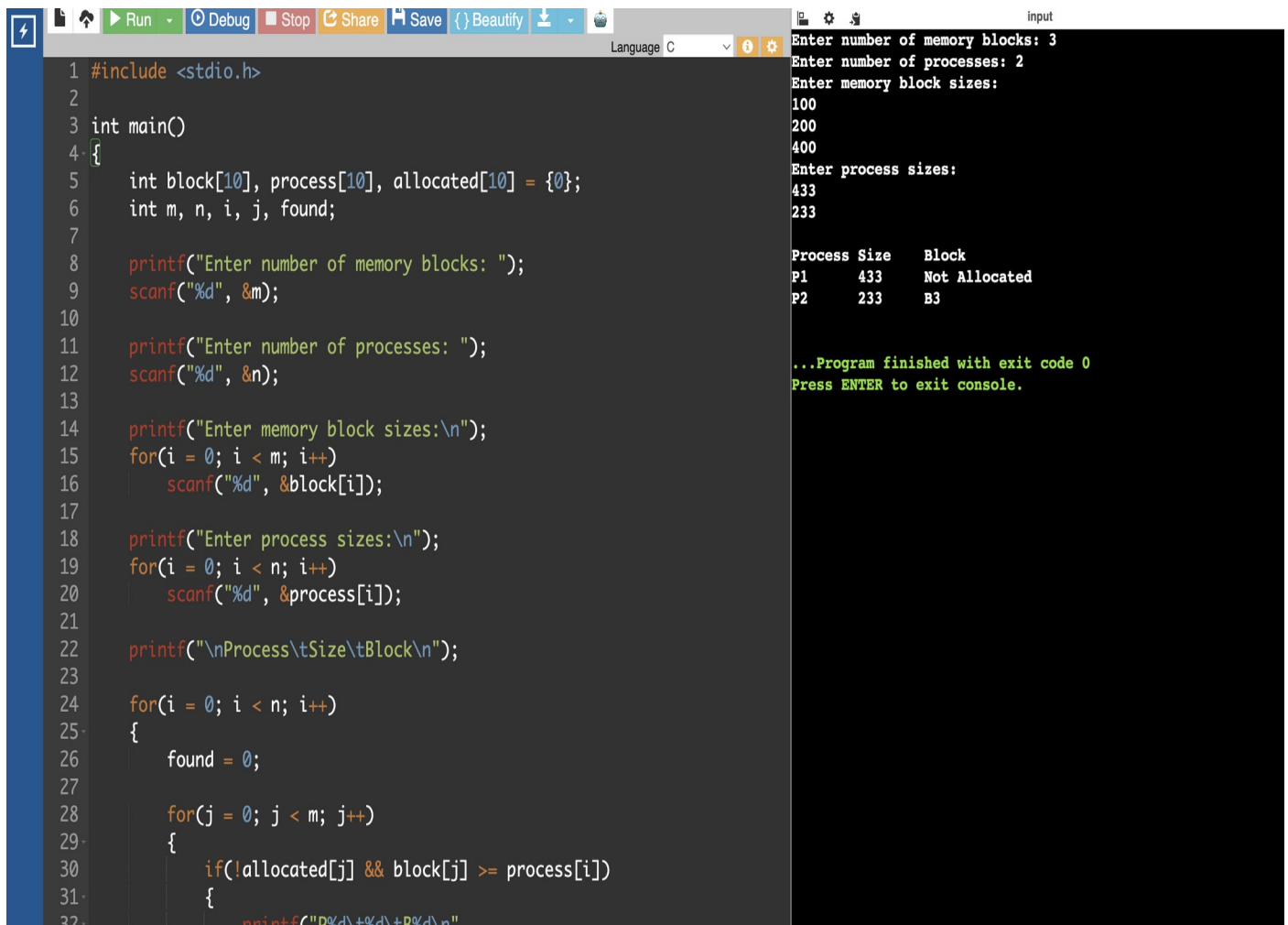


EXPERIMENT 23:

QUESTION:

Construct a C program to implement the first fit algorithm of memory management.

PROGRAM:



```
1 #include <stdio.h>
2
3 int main()
4 {
5     int block[10], process[10], allocated[10] = {0};
6     int m, n, i, j, found;
7
8     printf("Enter number of memory blocks: ");
9     scanf("%d", &m);
10
11     printf("Enter number of processes: ");
12     scanf("%d", &n);
13
14     printf("Enter memory block sizes:\n");
15     for(i = 0; i < m; i++)
16         scanf("%d", &block[i]);
17
18     printf("Enter process sizes:\n");
19     for(i = 0; i < n; i++)
20         scanf("%d", &process[i]);
21
22     printf("\nProcess\tSize\tBlock\n");
23
24     for(i = 0; i < n; i++)
25     {
26         found = 0;
27
28         for(j = 0; j < m; j++)
29         {
30             if(!allocated[j] && block[j] >= process[i])
31             {
32                 printf("P%d\t%d\tB%d\n", i, process[i], j);
33                 allocated[j] = 1;
34                 found = 1;
35             }
36         }
37
38         if(found == 0)
39             printf("P%d\t%d\tNot Allocated\n", i, process[i]);
40     }
41 }
```

Input

Enter number of memory blocks: 3
Enter number of processes: 2
Enter memory block sizes:
100
200
400
Enter process sizes:
433
233

Process	Size	Block
P1	433	Not Allocated
P2	233	B3

...Program finished with exit code 0
Press ENTER to exit console.